Name: _____

WORKSHEET

Quantifiers

1 Provide the mathematical symbol which represents each of these words/phrases.

a for all _____

b there exists _____

c element of _____

d real numbers _____

e rational numbers _____

f integers _____

g therefore _____

h implies _____

i not P _____

j required to prove _____

2 Rewrite each of these partial mathematical statements as word sentences.

a $\exists a \in \mathbb{R}$

b $\forall b \in \mathbb{Q}$

c $\forall c \in \mathbb{Z}$ such that $c < 0$

d $\exists d \in \mathbb{Q}^+$

e $\exists e \in \mathbb{Z}^-$ such that $e^2 = x$

f $\forall f \in \mathbb{R}, \exists y \in \mathbb{Z}$

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3 Use mathematical symbols to convert the following partial word sentences into mathematical statements.

a For all real numbers a

b b is greater than or equal to 10 but less than 36.

c There exists an integer c

d For all negative rational numbers d

e For all rational numbers e, there exists an integer x

f There exists integers f and g such that $w = \frac{g}{f}$

4 Rewrite each of these mathematical statements as word sentences.

a $\forall a \in \mathbb{Z}, \exists m \in \mathbb{R}$ such that $m = \sqrt{a}$

b $\forall b \in \mathbb{R}, \exists v, w \in \mathbb{R}$ such that $b = v - w$

c $\exists c \in \mathbb{Q}$ such that $0 < c \leq 20$

d $\forall d \in \mathbb{Q}, \exists s \in \mathbb{Z}$ such that $d = \frac{s}{s+1}$

e $\exists e \in \mathbb{Z}^+$ such that $\sqrt{e} \in \mathbb{R}$

f $\forall f \in \mathbb{R}^-, \exists x \in \mathbb{R}^+$ such that $x = f^4$

5 Use mathematical symbols to convert the following word sentences into mathematical statements.

a For all positive integers a, there exists a real number x such that $x = \sqrt{a}$.

b For all integers b, there exists a rational number y such that $y = \frac{1}{b}$.

c There exists a positive integer c such that the square root of c is also a positive integer.

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d There exists a real number d such that d is less than 0 but greater than or equal to -25.

e For all rational numbers e, there exist integers m and n such that $e = \frac{m^2}{n^3}$.

f For all negative rational numbers f, there exists a positive rational number z such that $f = -\frac{1}{z}$.

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Answers

1 **a** $\forall$

b $\exists$

c $\in$

d $\mathbb{R}$

e $\mathbb{Q}$

f $\mathbb{Z}$

g $\therefore$

h $\Rightarrow$

i P', $\neg P$ or $\overline{P}$

j RTP

2 **a** There exists a real number a.

b For all rational numbers b.

c For all negative integers c.

d There exists a positive rational number d.

e There exists a negative integer e such that $e^2 = x$.

f For all real numbers f, there exists an integer y.

3 **a** $\forall a \in \mathbb{R}$

b $10 \leq b < 36$

c $\exists c \in \mathbb{Z}$

d $\forall d \in \mathbb{Q}^-$ OR $\forall d \in \mathbb{Q}$ such that $d < 0$

e $\forall e \in \mathbb{Q}, \exists x \in \mathbb{Z}$

f $\exists f, g \in \mathbb{Z}$ such that $w = \frac{g}{f}$

4 **a** For all integers a, there exists a real number m such that $m = \sqrt{a}$.

b For all real numbers b, there exist real numbers v and w such that $b = v - w$.

c There exists a rational number c such that $0 < c \leq 20$.

d For all rational numbers d, there exists an integer s such that $d = \frac{s}{s+1}$

e There exists a positive integer e such that $\sqrt{e}$ is a real number.

f For all negative real numbers f, there exists a positive real number such that $x = f^4$.

5 **a** $\forall a \in \mathbb{Z}^+, \exists x \in \mathbb{R}$ such that $x = \sqrt{a}$ OR $\forall a \in \mathbb{Z}$ such that $a > 0$, $\exists x \in \mathbb{R}$ such that $x = \sqrt{a}$

b $\forall b \in \mathbb{Z}, \exists y \in \mathbb{Q}$ such that $y = \frac{1}{b}$

c $\exists c \in \mathbb{Z}^+$ such that $\sqrt{c} \in \mathbb{Z}^+$

d $\exists d \in \mathbb{R}$ such that $-25 \leq d < 0$

e $\forall e \in \mathbb{Q}, \exists m, n \in \mathbb{Z}$ such that $e = \frac{m^2}{n^3}$

f $\forall f \in \mathbb{Q}^-, \exists z \in \mathbb{Q}^+$ such that $f = -\frac{1}{z}$

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